

DEREK WANG

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Skills

Languages: C++, C, Java, Python, HTML5, CSS, JavaScript, TypeScript, SQL

Engineering Tools: Git, Fusion 360, Docker, Foxglove, PostgreSQL, MongoDB, Supabase

Technologies: React, FastAPI, Node.js, Express, ROS2, PyTorch, OpenCV, Ultralytics, Tailwind CSS

Certifications: Docker Foundations Professional

Experience

Autonomous Self Driving Software Engineer

Sept 2025 – Present

WATonomous

Waterloo, ON, Canada

- Enabled on-rover path following by designing and implementing a ROS 2 autonomy navigation stack featuring costmap generation, persistent map memory, an A* path planner, and a Pure Pursuit controller
- Improved object detection accuracy to 80% mAP from a 30% baseline by training a YOLO11s model on 3,448 images utilizing Ultralytics, learning rate scheduling, batch tuning, and data augmentation
- Achieved real-time model inference at 30 FPS on rover hardware by implementing a deployment pipeline utilizing ONNX export, onnxslim optimization, and embedded inference integration

Autonomy Developer

Aug 2025 – Present

Waterloo Aerial Robotics Group

Waterloo, ON, Canada

- Fulfilled autonomous flight testing requirements for the 2026 National Annual UAS Student Competition by developing Python-based Hardware-in-the-Loop (HITL) simulation systems using the MAVLink protocol
- Eliminated unrealistic teleportation behavior and created smooth trajectory interpolation by redesigning the GPS movement architecture using periodic functions for continuous position updates

Robotics Software Programmer – VEX Robotics Competition

Sept 2024 – June 2025

Checkmate Robotics Club

Markham, Ontario

- Reduced robot positioning error by 40% by programming competition software for Team 16868C Rushdown using Odometry positioning, PID Controllers, and comprehensive error handling
- Earned an invitation to the 2025 VRC Worlds (top 10% globally) and placed 5th out of 80 teams at Provincials by designing autonomous match routes and navigation algorithms featuring motion profiling and path planning

Projects

Silhouette | Node.js, Next.js, Express, MongoDB, Gemini, Overshoot.ai SDK, [Source Code](#)

- Secured a VC referral to build an active Proof of Concept for Merian Brothers, engineering a full-stack AI fashion advisor that processes live camera input with sub-500ms interaction latency.
- Implemented a Next.js + React frontend and Express.js backend with MongoDB Atlas, supporting 3+ style pipelines (Streetwear, Formal, Sporty) and dynamic product linking
- Orchestrated Overshoot.ai for live video outfit analysis with Gemini image/video models, designing prompt pipelines that manage async inference, rate limits, and style constraints to deliver personalized recommendations in under 2s

Coffee Chat Scheduler | React, Python, FastAPI, PostgreSQL, Google Calendar API, [Source Code](#)

- Built a full-stack web application using React and FastAPI, integrating Google Calendar API via OAuth 2.0 authentication with RESTful API endpoints, error handling, and CORS configuration
- Implemented real-time availability checking and automated calendar event creation, storing booking data in a PostgreSQL database with SQLAlchemy Object Relational Mapping, reducing booking confirmation time to under 1000ms

Autonomous Vehicle Control System | C++, ROS2, Docker, CMake, Foxglove, [Source Code](#)

- Engineered LiDAR-based perception pipeline processing sensor data to generate real-time costmaps with 0.1m resolution, integrating persistent memory mapping system for 95% accurate dynamic environment representation
- Architected a modular ROS2 navigation stack using A* pathfinding and Pure Pursuit control, achieving sub-0.5 m waypoint accuracy and 100% obstacle avoidance across 50+ scenarios.

Education

University of Waterloo

Expected Apr 2030

Bachelor of Applied Science in Computer Engineering

GPA: 3.7